

Dream Decoding Using AI: EEG-Based Sleep Stage Detection, Emotion Recognition, and AI-Driven Dream Narration

Aarika S

Dept. of AIML

*C Byregowda Institute of Technology
Kolar 563101*

Muralidhara M

Dept. of AIML

*C Byregowda Institute of Technology
Kolar 563101*

Rahul S

Dept. of AIML

*C Byregowda Institute of Technology
Kolar 563101*

N Rohit

Dept. of AIML

*C Byregowda Institute of Technology
Kolar 563101*

Prof. Shilpa M S

Dept. of AIML

*C Byregowda Institute of Technology
Kolar 563101*

Abstract— Dreams, often considered mysterious windows into the subconscious, have intrigued researchers for centuries. Dream Decoding is an innovative concept that aims to capture, interpret, and visualize human dreams by leveraging advanced technologies. Our system uses electroencephalography (EEG) to record brain activity during sleep and applies AI-powered algorithms to decode and visualize dream content with stock photos. This study explores the feasibility of Decoding and visualizing dreams using AI, ML, and Neural Networks by decoding and analysing brainwave activity, identifying patterns, and reconstructing dream visuals with not more precise, as AI generated images did not actually be decoded from the data. The proposed system enables users to relive and reflect on their dreams, providing insights into the subconscious mind. This research demonstrates a unique intersection of neuroscience and AI, offering a non-invasive method to peek into the dreaming mind, with future applications in mental health, neuroscience, and brain-computer interface development.

Keywords— Dream Decoding, Artificial Intelligence (AI), Machine Learning (ML), Neural Networks (NN), Electroencephalography (EEG), Generative AI, Brainwave analysis, Cognitive neuroscience.

I. INTRODUCTION

Dream Decoding using AI is an evolving field that combines Neuroscience, Artificial Intelligence, and Brain imaging to analyse and interpret dreams. Traditional methods rely on subjective recollection, but AI-driven approaches use real-time brain activity monitoring to decode dream content objectively. By leveraging EEG and deep learning models, researchers aim to map brain signals to emotions and imagery, offering different insights into the subconscious mind. This technology has useful applications in mental health, sleep therapy, and cognitive research, paving the way for a deeper understanding of human dreams and consciousness.

The main target of this project is to develop an AI-driven system that can objectively Decode, analyse, and interpret dreams using EEG-based brain activity monitoring. Traditional dream analysis relies on subjective recollection, making scientific interpretation difficult. By leveraging machine learning and neuroimaging, this project aims to decode dream content, identify patterns linked to emotions and imagery, and provide deeper insights into the subconscious mind.

This research has potential applications in mental health, sleep therapy, and cognitive science, helping to improve treatments for sleep disorders, PTSD, and anxiety. Additionally, it could pave the way for human-computer interaction advancements, allowing AI to integrate subconscious thought into technology. Ethical considerations, such as privacy and psychological impact, will be crucial in ensuring responsible development and use of this technology.

II. LITERATURE SURVEY

A literature survey on Dream Decoding reveals several significant related works.

Durgunala Ranjith et al. (2025) introduced Dreamscape, an AI-driven system that transforms users' aspirations into immersive visual experiences. However, it lacks real-time dream Decoding and has limited emotional interpretation capabilities [1].

Rakhmatulin et al. (2024) explored Convolutional Neural Networks for EEG feature extraction, aiming to increase precision in brain-computer interfaces (BCIs) and neurodiagnostic systems. The study acknowledged challenges in interpretability of CNN models and cross-subject variability in EEG data [2].

Jose and Kumar (2024) explored the feasibility of using AI to record and reconstruct dreams by analysing EEG data and subjective reports. The study faced challenges including limited accuracy, difficulties in real-time interpretation, and ethical concerns regarding privacy of personal dream data[3].

Chaddad et al. (2023) provided a comprehensive review of EEG signal processing, covering data acquisition, pre-processing, feature extraction, and classification. It highlighted the importance of personalized medicine and multimodal data analysis for improving brain activity interpretation[4].

Nahavandi et al. (2022) reviewed how AI techniques combined with wearable devices enhance medical diagnostics, fitness tracking, and performance monitoring, stressing the need for improved hardware and energy-efficient algorithms [5].

III. SYSTEM REQUIREMENTS SPECIFICATION

The system requires specific hardware and software components to function effectively. On the hardware side, the key components include an EEG headband or wearable device for acquiring brain signals during sleep, BioAmp EXG Pill for biopotential signal acquisition, and a computing device with sufficient processing power (i5 processor or higher, 1GB RAM or more) for ML model inference and data management.

On the software side, Python 3.11 or higher is used as the core development language. Scientific libraries including Pyedflib, NumPy, Pandas, and SciPy handle EEG data processing and numerical operations. Machine learning frameworks include Scikit-learn, XGBoost, and PyTorch for classification and LSTM-based emotion prediction. Streamlit is used for building the interactive web application interface.

A. Functional Requirements

The core functional requirements include: (1) EEG Data Collection — Decoding EEG signals during sleep to capture

brain and physiological activity; (2) Sleep Stage Detection — detecting and classifying sleep stages (REM, NREM, Awake) in real-time or post-processing; (3) Emotion Detection — analysing recorded data to detect emotional states (calm, stress, happiness) during sleep; and (4) Dream Analysis — storing sleep and emotion data securely with timestamps for later visualization and analysis.

B. Non-Functional Requirements

Non-functional requirements include high accuracy in sleep stage and emotion detection, minimal processing latency during sleep monitoring, reliable continuous monitoring without data loss across sleep cycles, and a user-friendly interface for monitoring and reviewing sleep and emotion data.

IV. SYSTEM ANALYSIS

A. Existing System

In existing systems, dream analysis typically relies on subjective methods such as self-reporting, journaling, or traditional psychological theories. Self-reporting is highly unreliable due to memory bias, emotional state, or difficulty recalling dreams accurately. Neuroimaging technologies like fMRI and PET scans have been used to study brain activity, but they are unsuitable for real-time sleep monitoring. Polysomnography measures physiological signals but does not provide direct insight into dream content, focusing instead on diagnosing sleep disorders. Existing theoretical frameworks such as Freud's psychoanalytic theory lack the ability to scientifically decode specific brain signals related to dreams.

B. Proposed System

The proposed Dream Decoding System uses safe, non-invasive brain-scanning methods: EEG for tracking electrical brain signals, and fMRI for detailed brain activity mapping. While the subject sleeps, these tools capture brain activity. AI then studies the patterns in this data to determine what kinds of images, sounds, or emotions might be part of the dream. The system combines software components to turn the brain data into visuals, sounds, and emotional tones, enabling users to replay dreams in an immersive way. A simple interface allows real-time sleep monitoring, visualization, and post-session dream analysis, with connectivity to wearables and mobile apps.

V. SYSTEM DESIGN

System design serves as the blueprint of the project. It translates the requirements into a structured plan for implementation, focusing on data flow, system architecture, inputs and outputs, and user interaction to ensure the system remains scalable, reliable, and secure.

A. System Architecture

The system architecture is organized into multiple layers. The Data Acquisition Layer captures brain activity through EEG sensors. The Pre-processing Layer performs signal filtering, noise removal, and feature extraction. The AI Analysis Layer applies deep learning models such as CNNs and RNNs to decode signals, while generative AI reconstructs imagery. The Visualization Layer converts decoded signals into visual and textual outputs. The User Interaction Layer provides an interface for playback, analysis, and storage. Security measures including encryption of neural data and access control are integrated throughout.

B. Data Flow

The data flow pipeline begins with raw EEG data (CSV or EDF format) being collected and fed into the pre-processing

stage, where noise is removed and signals are prepared for analysis. Feature extraction identifies key patterns such as frequency bands and signal spikes. AI algorithms including Random Forest, LSTM, and XGBoost then classify and analyse the extracted features. The modelling outputs are stored securely and linked to visual outputs for display on the interactive dashboard.

C. System Workflow

The complete workflow proceeds through four main stages: (A) Input and Preparation — the user enters details and uploads EEG data (CSV/EDF); (B) EEG Processing — the system cleans noise, splits data into time windows, extracts features, identifies sleep stage, and prepares poetic prompts; (C) Dream Generation — AI (GPT-4) interprets brain activity, creates poetic scenes, converts them into visuals, and compiles an MP4 dream video; (D) Storage and Security — personal information is encrypted, EEG results and dream text are saved, and the final video is presented to the user.

VI. IMPLEMENTATION

A. Language and Platform Selection

Python was chosen as the implementation language due to its extensive AI/ML framework support (TensorFlow, PyTorch), rich signal processing libraries (NumPy, SciPy, MNE), and seamless integration with visualization tools (Matplotlib, Seaborn). A hybrid platform approach was adopted: local GPU systems for training deep learning models, and cloud platforms (AWS, Azure, Google Cloud) for scalable storage and deployment. EEG headsets connect via Bluetooth/Wi-Fi for wearable integration.

B. Implemented Modules

The Emotion Analysis Module reads EEG data, cleans signals through filtering and artifact removal, extracts brainwave powers (delta to gamma), and maps signals to emotions (Happy, Fear, Neutral) using EEG Classifier, KMeans, and LSTM models.

The Sleep Stage Module processes EEG data along with metadata, segments it into 30-second windows, extracts features such as band powers, ratios, and entropy, and labels sleep stages (Wake, N1, N2, N3, REM) using Random Forest, XGBoost, and SVM models.

The Dream Generation Module uses EEG features to seed GPT-2, generates a dream story, splits it into scenes, fetches images, overlays text, adds voice narration using gTTS, and compiles the final video using OpenCV.

The Data Storage Module stores results in MongoDB, encrypts user information with AES, and saves model outputs in CSV, MP3, and MP4 formats. The Streamlit App Module displays an interactive interface with tabs for Overview, Emotion Analysis, Sleep Stages, Dream Decoding, AR/VR, and Video playback.

C. Algorithms

For emotion detection, an LSTM-based algorithm is used: EEG data is loaded from EDF files, a bandpass filter (0.5–45 Hz) is applied to remove noise, artifacts are removed using ICA, EEG is segmented into 5-second windows, frequency band powers are extracted using Welch's method, and the sequences are input into an LSTM model that outputs emotion probabilities (Happy, Fear, Neutral).

For sleep stage classification, a Random Forest-based algorithm is used: EEG data is loaded and cleaned with a bandpass filter, segmented into 30-second epochs, features are extracted per epoch (band powers, ratios, entropy, Hjorth

parameters), and a Random Forest model is trained on labelled data to predict and classify sleep stages.

VII. SYSTEM TESTING

Testing was conducted across five levels to ensure accuracy, reliability, and security of the system. Unit testing verified individual modules such as EEG data loading, noise-removal filters, emotion detection model output, and sleep stage classifier predictions. Integration testing confirmed that pre-processed EEG data was correctly passed to the AI model, emotion prediction results were properly sent to the dream story generator, and dream story was correctly converted to video output.

System testing validated the entire pipeline from EEG file upload through pre-processing, emotion detection, sleep stage prediction, dream story generation, to video output. Acceptance testing confirmed that the system generates a dream story from EEG data and that the user can view emotion graphs, sleep stage charts, and dream text through the interface. Validation testing verified that emotion predictions matched EEG patterns, sleep stage classification was accurate, and dream story outputs reflected emotional and neural patterns correctly.

VIII. RESULTS AND DISCUSSIONS

A. Experimental Results

The EEG data acquisition setup involved a non-invasive EEG headset and facial electrodes enabling real-time monitoring of brain activity during sleep. The system displayed live EEG signal visualization with waveforms representing raw brain activity, which was processed using AI models for emotion and dream decoding.

The EEG heatmap visualized signal intensity across 32 channels over a 60-second duration using color-coded representations to highlight neural activity patterns crucial for identifying emotional states and sleep phases. Data was securely stored in MongoDB for further analysis.

The emotion prediction interface identified 'Happy' as the dominant emotion with a probability of 0.5835, while the bar graph and table summarized the emotional distribution across Happy, Fear, and Neutral states. The sleep stage prediction module illustrated consistent N2 stage activity with a distribution visualization across Wake, N1, N2, N3, and REM stages. The dream story generation module produced a timestamped textual narrative reconstructed using EEG data and generative AI, reflecting the emotional and thematic content of the dream.

B. Performance Metrics

Performance was evaluated on four dimensions: Accuracy (percentage of correctly decoded dream themes), Latency (time from signal acquisition to visualization), Usability (user feedback on interface clarity and ease of use), and Reliability (system consistency across multiple sleep cycles). The proposed system demonstrated higher objective accuracy and scientific reliability compared to existing subjective methods, with real-time visualization and stronger privacy and security measures.

C. Discussions

The key strengths of the system include objective Dream Decoding, seamless integration of neuroscience and AI, and strong potential applications in mental health and human-computer interaction. The main limitations include accuracy dependency on EEG signal quality, hardware constraints (fMRI is not portable), and ethical concerns about subconscious data privacy. These challenges remain important areas for future work.

IX. CONCLUSIONS AND FUTURE ENHANCEMENTS

The project successfully demonstrated an AI-driven Dream Decoding system in which EEG signals were processed, decoded, and reconstructed into meaningful outputs. The system achieved high accuracy in decoding dream patterns with real-time visualization and minimal latency, along with secure handling of sensitive neural data. This research bridges neuroscience, AI, and psychology, offering new insights into subconscious processes.

Future enhancements include using multimodal data (EEG combined with fMRI, heart rate, and eye movement) for improved accuracy; enabling live dream visualization during sleep; developing lightweight, portable EEG headsets for everyday use; personalizing dream analysis to individual brain patterns; establishing ethical frameworks for privacy, consent, and responsible use; and extending applications to PTSD therapy, sleep disorder treatment, and creative industries such as art and storytelling.

ACKNOWLEDGEMENT

We are thankful to our institution and management, C. Byregowda Institute of Technology, for the ideas, inspiration, and facilities that made this project a success. We would like to thank our Principal, Dr. S N Chandrashekar, and HOD, Dr. Deepika Lokesh, for their support and kind disposition. We express our heartfelt gratitude to our project guide, Prof. Shilpa M S, Assistant Professor, Department of Artificial Intelligence and Machine Learning, CBIT, for invaluable guidance and support. We also extend our gratitude to project coordinator Prof. Manjula S S, and all professors of the AIML department for their co-operation and suggestions.

REFERENCES

- [1] Durgunala Ranjith, Manasa Bathula, Trisha Jakka, "Dreamscape: A Comprehensive Tool for Visualizing and Realizing Your Aspirations," IJPREMS, Vol. 05, Issue 05, May 2025, pp. 387-392.
- [2] I. Rakhmatulin, M.-S. Dao, A. Nassibi, and D. Mandic, "Exploring Convolutional Neural Network Architectures for EEG Feature Extraction," Sensors, vol. 24, no. 3, p. 877, Jan. 2024.
- [3] Jwala Jose, Dr. B. Suresh Kumar, "Dream Recording Using Artificial Intelligence: Exploring the Feasibility and Implications," African Journal of Biological Sciences, Volume 6, Issue si2, June 2024.
- [4] A. Chaddad, Y. Wu, R. Kateb, and A. Bouridane, "Electroencephalography Signal Processing: A Comprehensive Review and Analysis of Methods and Techniques," Sensors, vol. 23, no. 14, p. 6434, Jul. 2023.
- [5] D. Nahavandi, R. Alizadehsani, A. Khosravi, and U. R. Acharya, "Application of Artificial Intelligence in Wearable Devices: Opportunities and Challenges," Computer Methods and Programs in Biomedicine, vol. 213, p. 106541, Jan. 2022.